

MATHEMATICS
Introduction to Modern Algebra I
Final.

1) Determine all abelian groups of order 400.

$400 = 2^4 \times 5^2$. The possibilities are 1) $\mathbb{Z}_{16} \times \mathbb{Z}_{25}$, 2) $\mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_{25}$, 3) $\mathbb{Z}_4^2 \times \mathbb{Z}_{25}$, 4) $\mathbb{Z}_2^2 \times \mathbb{Z}_4 \times \mathbb{Z}_{25}$, 5) $\mathbb{Z}_2^2 \times \mathbb{Z}_{25}^2$, 6) $\mathbb{Z}_{16} \times \mathbb{Z}_5^2$, 7) $\mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_5^2$, 8) $\mathbb{Z}_4^2 \times \mathbb{Z}_5^2$, 9) $\mathbb{Z}_2^2 \times \mathbb{Z}_4 \times \mathbb{Z}_5^2$, 10) $\mathbb{Z}_2^4 \times \mathbb{Z}_5^2$.

2) To which of the following groups is $\mathbb{Z}_{20} \times \mathbb{Z}_{50} \times \mathbb{Z}_2$ isomorphic to? $\mathbb{Z}_2^2 \times \mathbb{Z}_4 \times \mathbb{Z}_5^3$, $\mathbb{Z}_{2000}$, $\mathbb{Z}_2^2 \times \mathbb{Z}_4 \times \mathbb{Z}_5 \times \mathbb{Z}_{25}$, or $\mathbb{Z}_4^2 \times \mathbb{Z}_5 \times \mathbb{Z}_{25}$?

$\mathbb{Z}_{20} \times \mathbb{Z}_{50} \times \mathbb{Z}_2 \cong \mathbb{Z}_4 \times \mathbb{Z}_5 \times \mathbb{Z}_2 \times \mathbb{Z}_{25} \times \mathbb{Z}_2 \cong \mathbb{Z}_2^2 \times \mathbb{Z}_4 \times \mathbb{Z}_5 \times \mathbb{Z}_{25}$.

3) Prove that there is no surjective homomorphism $\phi : \mathbb{Z}_4^2 \rightarrow \mathbb{Z}_8$. Find a surjective homomorphism $\psi : D_4 \rightarrow \mathbb{Z}_2^2$ (Hint: find a normal subgroup N of index 4 and show that $G/N \cong \mathbb{Z}_2^2$.)

$\mathbb{Z}_8$ is cyclic and so the image of $\mathbb{Z}_4^2$ is cyclic hence generated by an element say $\phi(x)$ of order 8. Since every element $x \in \mathbb{Z}_4^2$ has order ≤ 4 , this is impossible.

Let $N = Z(G)$, then $|Z(G)| = 2$ and in fact $Z(G) = \{e, \rho^2\}$. We must show that $D_4/Z(G) \cong \mathbb{Z}_2^2$ i.e. that elements in $D_4/Z(G)$ have order at most 2. Let $Z = Z(G)$. We have $D_4/Z = \{Z, \rho Z, \sigma Z, \rho\sigma Z\}$. Clearly $(\rho Z)^2 = \rho^2 Z = Z$, $(\sigma Z)^2 = \sigma^2 Z = Z$ and $(\rho\sigma Z)^2 = \rho\sigma\rho\sigma Z = \rho\rho^3 Z = Z$.

4) Write $(1, 7, 3)(2, 3, 4)(5, 8, 6)(8, 9)$ as a product of disjoint cycles **and** as a product of transpositions. What is its order **and** sign?

$$\begin{aligned} (1, 7, 3)(2, 3, 4)(5, 8, 6)(8, 9) &= (1, 7, 3, 4, 2)(5, 8, 9, 6) = \\ &= (1, 2)(1, 4)(1, 3)(1, 7)(5, 6)(5, 9)(5, 8). \end{aligned}$$

The order is 20 and the sign is odd.

5) Define U_m , prove that it is a group and show that if a and m are two positive integers such that $(a, m) = 1$ then $a^{\varphi(m)} = 1$ modulo m .

We let $U_m = \{[x] | 0 < x < m, \text{ and } (x, m) = 1\}$ where $[x]$ denotes the equivalence class of x modulo m . U_m is a group with respect to multiplication. It is closed under multiplication since $(x, m) = 1$ and $(y, m) = 1$ implies $(xym) = 1$. The identity element is $[1]$, associativity follows from associativity in $\mathbb{Z}$: $([a] \cdot [b]) \cdot [c] = [a \cdot b] \cdot [c] = [(a \cdot b) \cdot c] = [a \cdot (b \cdot c)] = [a] \cdot [b \cdot c] = [a] \cdot ([b] \cdot [c])$. To prove the existence of the inverse of $[x] \in U_m$, note that as $(x, m) = 1$, we can write $ax + bm = 1$ for some integers a, b . But then $(a, m) = 1$ so that $[a] \in U_m$ and $[x][a] = [xa] = [xa + bm] = [1]$ as required.

Since $|U_m| = \varphi(m)$, the order of any element $[a] \in U_m$ divides $\varphi(m)$. But then $[a^{\varphi(m)}] = [a]^{\varphi(m)} = [1]$ as required.

6) Let G be a group of order p^m where p is prime. Show that $Z(G) \neq \{e\}$.

By the Class Equation we have

$$G = \sum_{a \in G} cl(a) = \sum_{a \in G} \frac{|G|}{|C(a)|} = |Z(G)| + \sum_{a \in G - Z(G)} \frac{|G|}{|C(a)|}.$$

Note that p divides G and $\frac{|G|}{|C(a)|}$ for any $a \in G - Z(G)$. Therefore p divides $|Z(G)|$. Since $e \in Z(G)$, we have $|Z(G)| \geq p$ so that $Z(G) \neq \{e\}$.

7) Let p be a prime. Prove that any group of order p^2 is abelian (you may use ex. 6).

We know that $|Z(G)| \in \{1, p, p^2\}$. By ex. 6) $|Z(G)| \neq 1$. If $|Z(G)| = p^2$, then G is abelian and so we may assume that $|Z(G)| = p$. Let $x \in G - Z(G)$ and consider the subgroup $C(x)$. We have $Z(G) \subset C(x)$ and $x \in C(x)$ so that $|C(x)| > p$. Since $|C(x)|$ divides p^2 , we have $|C(x)| = p^2$ and so $C(x) = G$ i.e. $x \in Z(G)$ which is impossible.

8) If M and N are normal subgroups of a group G , such that $M \cap N = \{e\}$ and $MN = G$, then show that $mn = nm$ for any $n \in N$ and $m \in M$ and that $M \times N$ is isomorphic to G .

Let $n \in N$ and $m \in M$. As N, M are normal subgroups, $mn^{-1}m^{-1} \in N$, and $nmm^{-1} \in M$. But then $nmm^{-1}m^{-1} = n(mn^{-1}m^{-1}) \in N$ and $nmm^{-1}m^{-1} = (nmm^{-1})m^{-1} \in M$ so that $nmm^{-1}m^{-1} \in M \cap N = \{e\}$. Thus $nmm^{-1}m^{-1} = e$ and so $nm = mn$.

Define $\phi : M \times N \rightarrow G$ by $\phi(n, m) = n \cdot m$. Then ϕ is a homomorphism because

$$\phi((n, m)(n', m')) = \phi(nn', mm') = nn'mm' = nmn'm' = \phi(n, m)\phi(n', m').$$

ϕ is injective because $\phi(n, m) = e$ implies $nm = e$ so that $n = m^{-1} \in N \cap M = \{e\}$. Thus $n = m^{-1} = e$ i.e. $(n, m) = (e, e)$ is the identity.

ϕ is surjective because (as $G = NM$) for any $g \in G$ we have $g = nm = \phi(n, m)$.

9) State and prove Lagrange's Theorem.

Theorem. Let H be a subgroup of a finite group G , then the order of H divides the order of G .

Proof. Define the equivalence relation $a \sim b$ iff $ab^{-1} \in H$. (Symmetric: $a \sim a$ as $aa^{-1} = e \in H$; Reflexive: $a \sim b$ implies $ab^{-1} \in H$ so that $ba^{-1} = (ab^{-1})^{-1} \in H$ and hence $b \sim a$; Transitive: if $a \sim b$ and $b \sim c$, then $ab^{-1} \in H$ and $bc^{-1} \in H$ so that $ac^{-1} = ab^{-1}bc^{-1} \in H$. Thus $a \sim c$.)

It is easy to see that $[a] = Ha$ and that $|Ha| = |H|$ for any $a \in G$. But then since $G = \cup_{a \in G} Ha$, we have that the order of G is the product of $|H|$ times the number of equivalence classes. $\square$

10) State and prove Cayley's Theorem.

Theorem. Let G be any group, then there is a set S such that G is a subgroup of $A(S)$.

Proof. Let $S = G$ and define $\phi : G \rightarrow S$ by $\phi(g) = \sigma_g$ where $\sigma_g(x) = gx$.

We check that ϕ is a homomorphism i.e. that $\phi(gg') = \phi(g) \circ \phi(g')$ for all $g, g' \in G$, i.e. that $\phi(gg')(x) = (\phi(g) \circ \phi(g'))(x)$ for all $x \in S$. In fact $\phi(gg')(x) = \sigma_{gg'}(x) = (gg')x = g(g'x) = \sigma_g(\sigma_{g'}(x)) = \phi(g)(\phi(g')(x)) = (\phi(g) \circ \phi(g'))(x)$.

Finally we check that ϕ is injective. Suppose that $\phi(g) = id$, then $\phi(g)(e) = id(e)$ so that $g = ge = e$. $\square$